

EXTRACTING SALT

SEPARATING MIXTURES

The substance we call **table salt** has many uses, not just flavoring foods. It is mostly made of a mineral called sodium chloride, widely found on Earth but often dissolved in water. This cannot be filtered out, but it can be separated by evaporation. Where salty water is trapped in shallow pools, the heat of the Sun causes the water to evaporate, leaving behind piles of white salt.

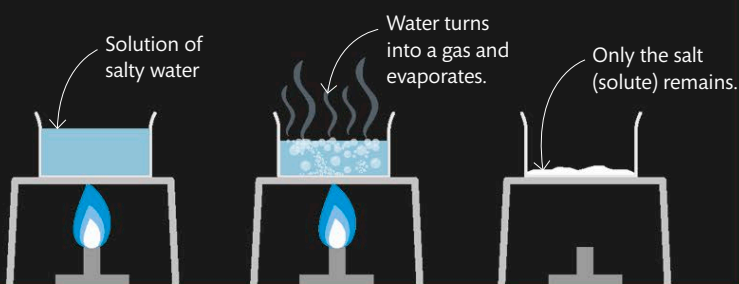
WINTER ROADS

Salt is spread on roads in winter, where it mixes with rain and snow to form salty water. This has a lower melting point than pure water, so it stays liquid at colder temperatures. The roads are kept free of ice, making them less slippery and safer for drivers.



EVAPORATION

Salty water is a solution—a mixture in which one substance (the solute) is dissolved in another (the solvent). The salt is known as the solute. We can separate the solute from the solution by evaporation. The liquid water turns into a gas (water vapor), leaving solid salt particles behind.



1. The salty water is heated gently.

2. As the water evaporates, the solution becomes more concentrated and salty.

3. When all the water has evaporated, only solid salt crystals are left.

Salt flats

Salt can be naturally extracted from large, flat areas of land called salt flats. The world's largest salt flat is Salar de Uyuni in Bolivia, an ancient lake that covers more than 3,900 sq miles (10,000 sq km). Salar de Uyuni is also a source of lithium, magnesium, and potassium.

